Corrine Sadler

D598-Analytics Programming

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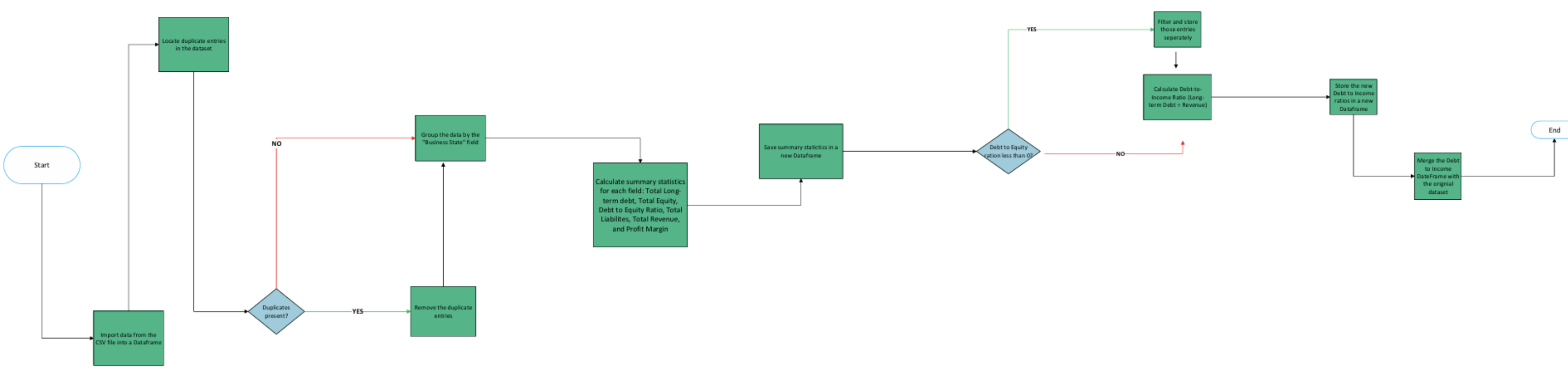
Task 1: Program Planning

A. Flowchart



B. Pseudocode

1. Begin
2. Import Data from the CSV file into a Dataframe
3. Locate duplicate entries in the dataset
4. If duplicates exist:
    a. Remove the duplicate entries
       OR
    b. Continue the dataset as-is
5. Group the data by the “Business State” field
6. For each of the following groups:
    a. Total Long-term Debt
    b. Total Equity
    c. Debt-to-Equity Ratio
    d. Total Liabilities
    e. Total Revenue
    f. Profit Margin

Calculate the following:

- Mean
- Median
- Minimum
- Maximum

7. Save the results in a new data frame
8. Identify if there are any values in the Debt-to-Equity Ratio that are less than 0
    a. If YES, filter and store those entries separately
    b. If NO, continue the dataset as-is
9. Calculate the Debt-to-Income ratio (Total Debt/ Total Revenue) for each entry
10. Concatenate Debt-to-Income with Original Data
11. End

C.

1. Flowchart Explanation

The flowchart and pseudocode were designed to systematically analyze business performance data from the D598 Dataset logically and efficiently. The process begins by importing the dataset into a data frame to ensure the data is available for analysis. Duplicate entries are then identified and removed, if present, to maintain data accuracy and prevent duplicated records from skewing the analysis results.

Once the dataset is cleaned, the data is grouped by the “Business State” field to allow for state-level analysis. Descriptive statistics, including the mean, median, minimum, and maximum, are calculated for all relevant numeric variables such as total long-term debt, debt-to-equity ratio, total liabilities, total revenue, and profit margin. These results are stored in a new data frame to preserve the original dataset and support further analysis.

The program then evaluates the Debt-to-Equity column to identify businesses with negative ratios. A conditional check is used to filter and store these businesses separately, as a negative debt-to-income ratio may indicate potential financial risk. After this step, the program calculates debt-to-income ratios for each company by dividing total long-term debt by total revenue. The calculated debt-to-income ratios are stored in a new data frame and then merged with the original dataset to create a final data frame that includes both the original financial data and the required debt-to-income ratio. The final "End" node in the flowchart provides a clear visual summary of the program's logic, while the pseudocode presents the same reasoning in detailed, step-by-step instructions. This alignment ensures the analytical process is easy to follow and meets the rubric's expectations for clarity, accuracy, and consistency.

2. Pseudocode Explanation

The pseudocode outlines the logical steps required to analyze the D598 Data Set in a structured and sequential manner. The step begins by importing the dataset into a data frame to ensure all business records are available for processing. The pseudocode then includes a conditional check to identify and remove duplicate rows, which helps maintain data integrity and ensures that calculations are based on unique records only.

After the data is cleaned, the pseudocode groups the data by the "Business State" field and calculates descriptive statistics for each group. These statistics include the mean, median, minimum, and maximum values for all required numeric variables, such as total long-term debt, total equity, debt-to-equity ratio, total liabilities, total revenue, and profit margin. The results of these calculations are stored in a separate data frame to preserve the original dataset for subsequent analysis.

Next, the pseudocode evaluates the Debt-to-Equity column to determine whether any businesses have negative ratios. A conditional statement is used to filter and store these businesses in a separate data frame when such values are present. This step enables the identification of potentially high-risk businesses without interrupting the remainder of the analytical process.

The pseudocode then calculates the debt-to-income ratio for each business by dividing total long-term debt by total revenue. These calculated ratios are stored in a new data frame and merged with the original dataset using the business identifier as the key. This final step produces a consolidated data frame that includes both the original financial variables and the required debt-to-income ratio. The pseudocode concludes with an explicit termination step, clearly defining the end of the program's execution.

References

The only sources used were the official course materials from WGU.